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JEE MAIN · QUESTION PAPER

JEE Main 2022 (Online) 27th July Evening Shift

JEE Main Question Paper with Answer Key

27 Jul 2022	90	25	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

Chemistry	21 questions
Mathematics	19 questions
Physics	25 questions
Out of syllabus	25 questions
Answer key	at the end

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Chemistry 21 questions

Q1. MCQ (Single Correct Answer)

The correct decreasing order of energy for the orbitals having, following set of quantum numbers :

(A) n = 3, l = 0, m = 0

(B) n = 4, l = 0, m = 0

(C) n = 3, l = 1, m = 0

(D) n = 3, l = 2, m = 1

is :

(A) (D) > (B) > (C) > (A)

(B) (B) > (D) > (C) > (A)

(C) (C) > (B) > (D) > (A)

(D) (B) > (C) > (D) > (A)

Open solution

Q2. MCQ (Single Correct Answer)

Match List - I with List - II.

	List - I		List - II
(A)	$\psi_{MO} = \psi_A - \psi_B$	(I)	Dipole moment
(B)	$\mu = Q \times r$	(II)	Bonding molecular orbital
(C)	$\frac{N_b - N_a}{2}$	(III)	Anti-bonding molecular orbital
(D)	$\psi_{MO} = \psi_A + \psi_B$	(IV)	Bond order

Choose the correct answer from the options given below :

(A) $(A) - (II), (B) - (I), (C) - (IV), (D) - (III)$

(B) $(A) - (III), (B) - (IV), (C) - (I), (D) - (II)$

(C) $(A) - (III), (B) - (I), (C) - (IV), (D) - (II)$

(D) $(A) - (III), (B) - (IV), (C) - (II), (D) - (I)$

Open solution

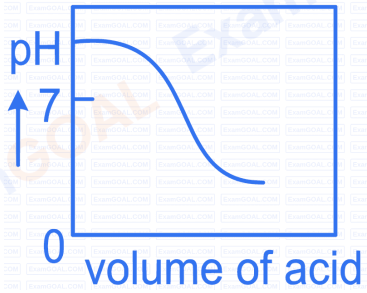
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Q3. MCQ (Single Correct Answer)

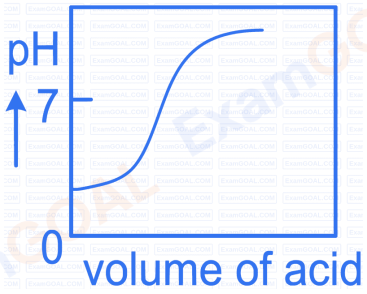
Open solution

The plot of pH-metric titration of weak base NH_4OH vs strong acid HCl looks like :

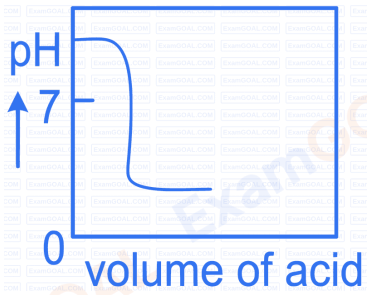
(A)



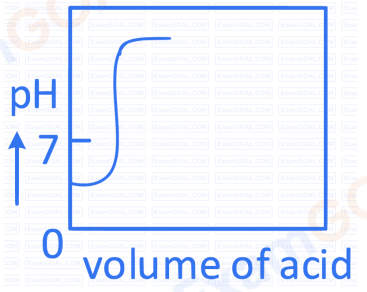
(B)



(C)



(D)



Q4. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement I : For KI, molar conductivity increases steeply with dilution

Statement II : For carbonic acid, molar conductivity increases slowly with dilution

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

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Q5. MCQ (Single Correct Answer)

Outermost electronic configurations of four elements A, B, C, D are given below :

(A) $3s^2$

(B) $3s^2 3p^1$

(C) $3s^2 3p^3$

(D) $3s^2 3p^4$

The correct order of first ionization enthalpy for them is :

(A) (A) < (B) < (C) < (D)

(B) (B) < (A) < (D) < (C)

(C) (B) < (D) < (A) < (C)

(D) (B) < (A) < (C) < (D)

Open solution

Q6. MCQ (Single Correct Answer)

In neutral or alkaline solution, MnO_4^- oxidises thiosulphate to :

(A) $S_2O_7^{2-}$

(B) $S_2O_8^{2-}$

(C) SO_3^{2-}

(D) SO_4^{2-}

Open solution

Q7. MCQ (Single Correct Answer)

Low oxidation state of metals in their complexes are common when ligands :

(A) have good π-accepting character

(B) have good σ-donor character

(C) are having good π-donating ability

(D) are having poor σ-donating ability

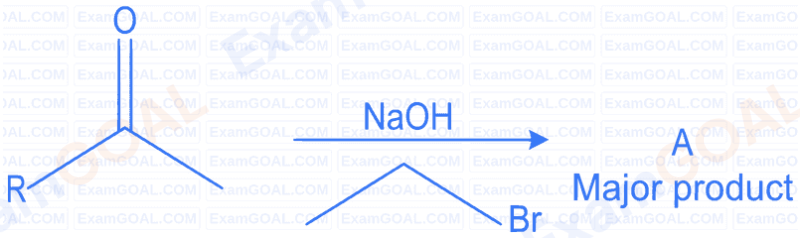
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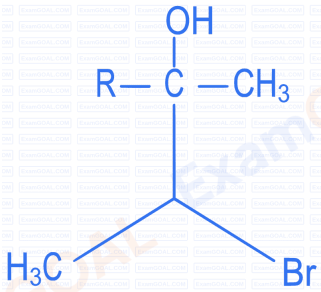
Q8. MCQ (Single Correct Answer)

Open solution

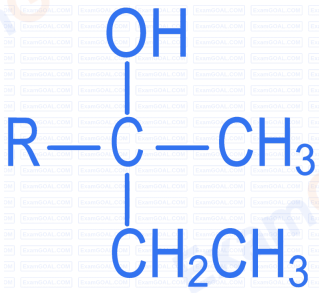
The structure of A in the given reaction is :



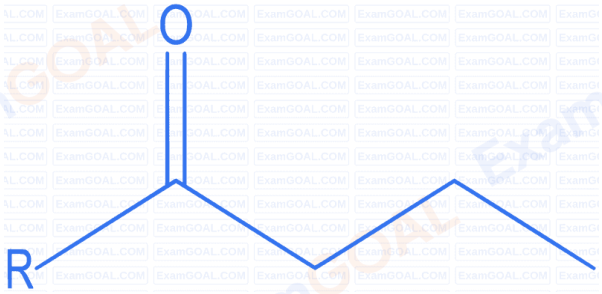
(A)



(B)



(C)



(D)

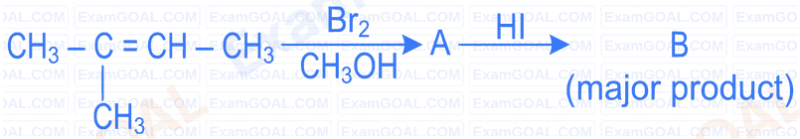


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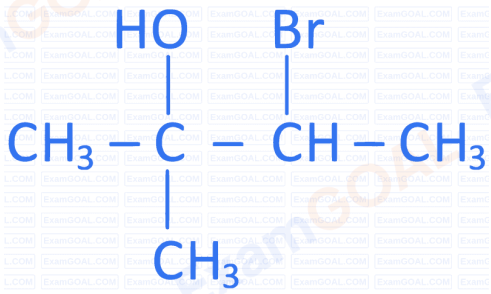
Q9. MCQ (Single Correct Answer)

Open solution

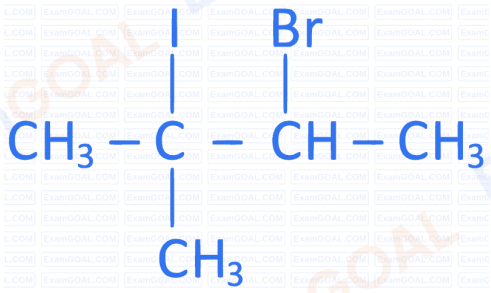
Major product 'B' of the following reaction sequence is :



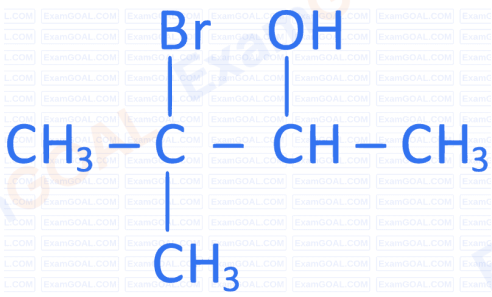
(A)



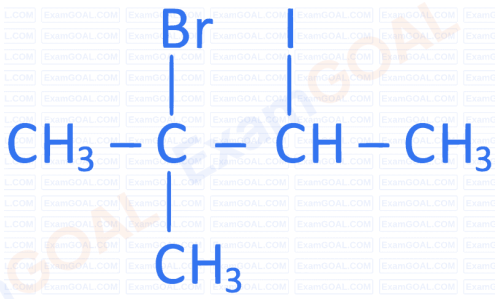
(B)



(C)



(D)

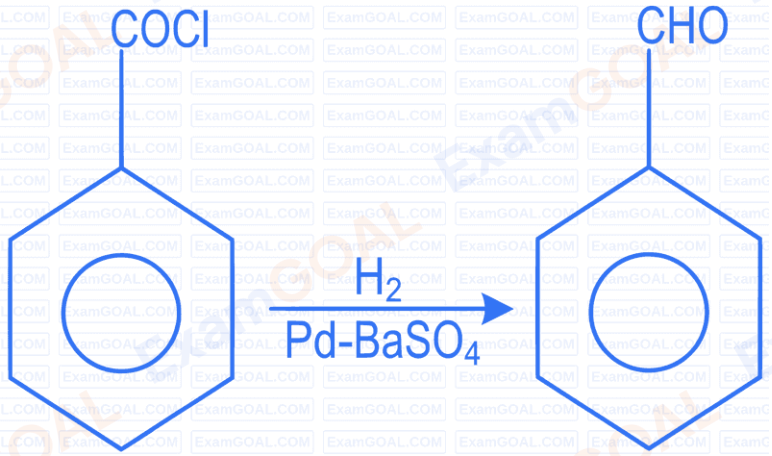
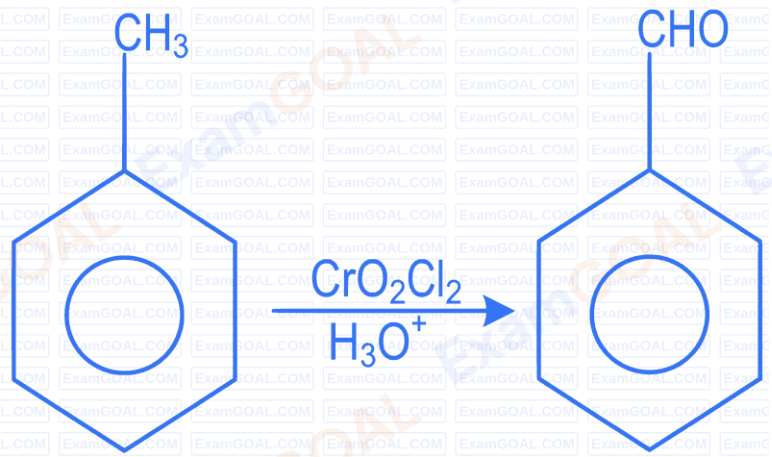
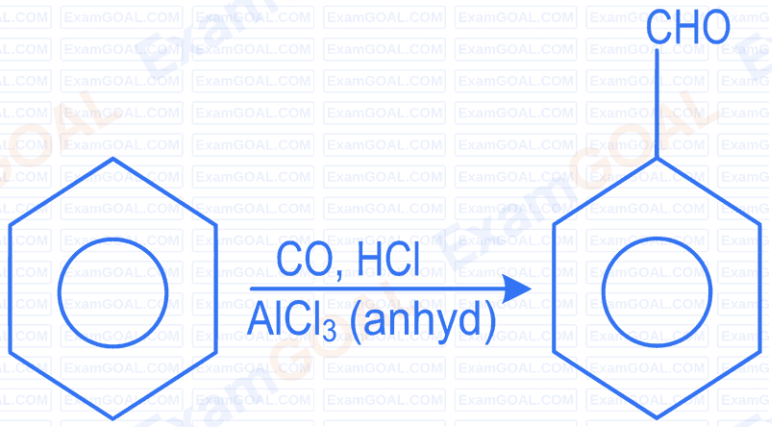


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Q10. MCQ (Single Correct Answer)

Open solution

Match List - I with List - II.

	List - I		List - II
(A)		(I)	Gatterman Koch reaction
(B)	$CH_3 - CN \xrightarrow[H_3O^+]{SnCl_2/HCl} CH_3 - CHO$	(II)	Etard reaction
(C)		(III)	Stephen reaction
(D)		(IV)	Rosenmund reaction

Choose the correct answer from the options given below :

(A) $(A) - (IV), (B) - (III), (C) - (II), (D) - (I)$

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(B) $(A)-(I),(B)-(II),(C)-(III),(D)-(IV)$

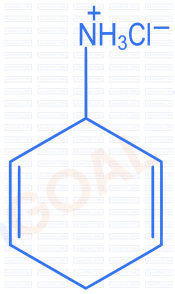
(C) $(A)-(II),(B)-(III),(C)-(IV),(D)-(I)$

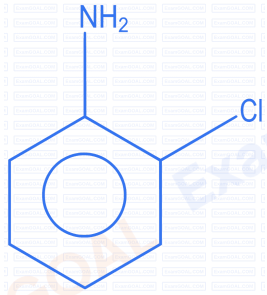
(D) $(A)-(III),(B)-(II),(C)-(I),(D)-(IV)$

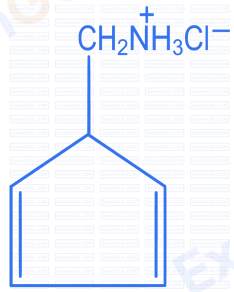
Q11. MCQ (Single Correct Answer)

Open solution

An organic compound $'A'$ contains nitrogen and chlorine. It dissolves readily in water to give a solution that turns litmus red. Titration of compound $'A'$ with standard base indicates that the molecular weight of $'A'$ is 131 ± 2. When a sample of $'A'$ is treated with aq. $NaOH$, a liquid separates which contains N but not Cl. Treatment of the obtained liquid with nitrous acid followed by phenol gives orange precipitate. The compound $'A'$ is :

(A) 

(B) 

(C) 

(D) 

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Q12. MCQ (Single Correct Answer)

Open solution

Match List - I with Match List - II.

	List - I		List - II
(A)	Glucose + HI	(I)	Gluconic acid
(B)	Glucose + Br_2 water	(II)	Glucose pentacetate
(C)	Glucose + acetic anhydride	(III)	Saccharic acid
(D)	Glucose + HNO_3	(IV)	Hexane

Choose the correct answer from the options given below:

(A) (A) - (IV), (B) - (I), (C) -(II), (D) - (III)

(B) (A) - (IV), (B) - (III), (C) -(II), (D) - (I)

(C) (A) - (III), (B) - (I), (C) -(IV), (D) - (II)

(D) (A) - (I), (B) - (III), (C) -(IV), (D) - (II)

Q13. MCQ (Single Correct Answer)

Open solution

Match List - I with List - II.

	List - I (Mixture)		List - II (Purification Process)
(A)	Chloroform & Aniline	(I)	Steam distillation
(B)	Benzoic acid & Naphthalene	(II)	Sublimation
(C)	Water & Aniline	(III)	Distillation
(D)	Naphthalene & Sodium chloride	(IV)	Crystallisation

Choose the correct answer from the options given below :

(A) (A) - (IV), (B) - (III), (C) - (I), (D) - (II)

(B) (A) - (III), (B) - (I), (C) - (IV), (D) - (II)

(C) (A) - (III), (B) - (IV), (C) - (II), (D) - (I)

(D) (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Q14. MCQ (Single Correct Answer)

Open solution

Fe^{3+} cation gives a prussian blue precipitate on addition of potassium ferrocyanide solution due to the formation of :

(A) $[Fe(H_2O)_6]_2\,[Fe(CN)_6]$

(B) $Fe_2[Fe(CN)_6]_2$

(C) $Fe_3[Fe(OH)_2(CN)_4]_2$

(D) $Fe_4[Fe(CN)_6]_3$

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Q15. Numerical

The normality of H_2SO_4 in the solution obtained on mixing $100\ mL$ of $0.1\ M\ H_2SO_4$ with $50\ mL$ of $0.1\ M\ NaOH$ is _____ $\times 10^{-1}$ N. (Nearest Integer)

Open solution

Q16. Numerical

A gas (Molar mass = 280 $g\ mol^{-1}$) was burnt in excess O_2 in a constant volume calorimeter and during combustion the temperature of calorimeter increased from $298.0\ K$ to $298.45\ K$. If the heat capacity of calorimeter is $2.5\ kJ\ K^{-1}$ and enthalpy of combustion of gas is $9\ kJ\ mol^{-1}$ then amount of gas burnt is _____ g. (Nearest Integer)

Open solution

Q17. Numerical

When a certain amount of solid A is dissolved in $100\ g$ of water at $25^{\circ}C$ to make a dilute solution, the vapour pressure of the solution is reduced to one-half of that of pure water. The vapour pressure of pure water is $23.76\ mmHg$. The number of moles of solute A added is _____. (Nearest Integer)

Open solution

Q18. Numerical

$$\underset{\text{Reactant}}{[A]} \rightarrow \underset{\text{Product}}{[B]}$$

If formation of compound [B] follows the first order of kinetics and after 70 minutes the concentration of [A] was found to be half of its initial concentration. Then the rate constant of the reaction is $x \times 10^{-6}\ s^{-1}$. The value of x is _____. (Nearest Integer)

Open solution

Q19. Numerical

The number of molecule(s) or ion(s) from the following having non-planar structure is _____.

$NO_3^-, H_2O_2, BF_3, PCl_3, XeF_4, SF_4, XeO_3, PH_4^+, SO_3, [Al(OH)_4]^-$

Open solution

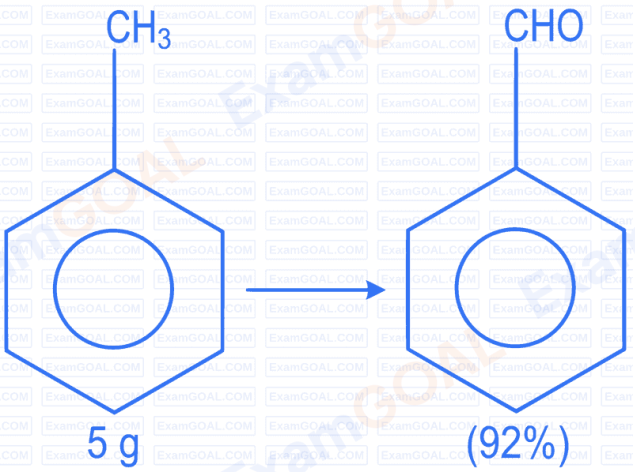
Q20. Numerical

The spin only magnetic moment of the complex present in Fehling's reagent is _____ B.M. (Nearest integer).

Open solution

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Q21. Numerical



Open solution

In the above reaction, 5 g of toluene is converted into benzaldehyde with 92% yield. The amount of benzaldehyde produced is _____ $\times 10^{-2}$ g. (Nearest integer)

Mathematics 19 questions

Q22. MCQ (Single Correct Answer)

The domain of the function $f(x) = \sin^{-1}\left[2x^2 - 3\right] + \log_2\left(\log_{\frac{1}{2}}\left(x^2 - 5x + 5\right)\right)$, where [t] is the greatest integer function, is :

(A) $\left(-\sqrt{\frac{5}{2}}, \frac{5-\sqrt{5}}{2}\right)$

(B) $\left(\frac{5-\sqrt{5}}{2}, \frac{5+\sqrt{5}}{2}\right)$

(C) $\left(1, \frac{5-\sqrt{5}}{2}\right)$

(D) $\left[1, \frac{5+\sqrt{5}}{2}\right)$

Open solution

Q23. MCQ (Single Correct Answer)

Let S be the set of all $(\alpha, \beta), \pi < \alpha, \beta < 2\pi$, for which the complex number $\frac{1-i\sin\alpha}{1+2i\sin\alpha}$ is purely imaginary and $\frac{1+i\cos\beta}{1-2i\cos\beta}$ is purely real. Let $Z_{\alpha\beta} = \sin 2\alpha + i\cos 2\beta, (\alpha, \beta) \in S$. Then $\sum\limits_{(\alpha,\beta)\in S}\left(iZ_{\alpha\beta} + \frac{1}{i\bar{Z}_{\alpha\beta}}\right)$ is equal to :

(A) 3

(B) 3 i

(C) 1

(D) $2 - i$

Open solution

Q24. MCQ (Single Correct Answer)

If α, β are the roots of the equation

$x^2 - \left(5 + 3^{\sqrt{\log_3 5}} - 5^{\sqrt{\log_5 3}}\right)x + 3\left(3^{(\log_3 5)^{\frac{1}{3}}} - 5^{(\log_5 3)^{\frac{2}{3}}} - 1\right) = 0,$

then the equation, whose roots are $\alpha + \frac{1}{\beta}$ and $\beta + \frac{1}{\alpha}$, is :

(A) $3x^2 - 20x - 12 = 0$

(B) $3x^2 - 10x - 4 = 0$

(C) $3x^2 - 10x + 2 = 0$

(D) $3x^2 - 20x + 16 = 0$

Open solution

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Q25. MCQ (Single Correct Answer)

If for $p \neq q \neq 0$, the function $f(x) = \frac{\sqrt[7]{p(729+x)}-3}{\sqrt[3]{729+qx}-9}$ is continuous at $x = 0$, then :

(A) $7pq\,f(0) - 1 = 0$ (B) $63q\,f(0) - p^2 = 0$ (C) $21q\,f(0) - p^2 = 0$ (D) $7pq\,f(0) - 9 = 0$

Open solution

Q26. MCQ (Single Correct Answer)

Let $f(x) = 2 + |x| - |x-1| + |x+1|, x \in \mathbf{R}$.

Consider

$(S1) : f'\left(-\frac{3}{2}\right) + f'\left(-\frac{1}{2}\right) + f'\left(\frac{1}{2}\right) + f'\left(\frac{3}{2}\right) = 2$

$(S2) : \int_{-2}^{2} f(x)dx = 12$

Then,

(A) both (S1) and (S2) are correct (B) both (S1) and (S2) are wrong

(C) only (S1) is correct (D) only (S2) is correct

Open solution

Q27. MCQ (Single Correct Answer)

Let the sum of an infinite G.P., whose first term is a and the common ratio is r, be 5 . Let the sum of its first five terms be $\frac{98}{25}$. Then the sum of the first 21 terms of an AP, whose first term is $10ar$, n^{th} term is a_n and the common difference is $10ar^2$, is equal to :

(A) $21\,a_{11}$ (B) $22\,a_{11}$ (C) $15\,a_{16}$ (D) $14\,a_{16}$

Open solution

Q28. MCQ (Single Correct Answer)

The area of the region enclosed by $y \leq 4x^2, x^2 \leq 9y$ and $y \leq 4$, is equal to :

(A) $\frac{40}{3}$ (B) $\frac{56}{3}$ (C) $\frac{112}{3}$ (D) $\frac{80}{3}$

Open solution

Q29. MCQ (Single Correct Answer)

$\int_0^2 \left(\left|2x^2 - 3x\right| + \left[x - \frac{1}{2}\right]\right)dx$, where [t] is the greatest integer function, is equal to :

(A) $\frac{7}{6}$ (B) $\frac{19}{12}$ (C) $\frac{31}{12}$ (D) $\frac{3}{2}$

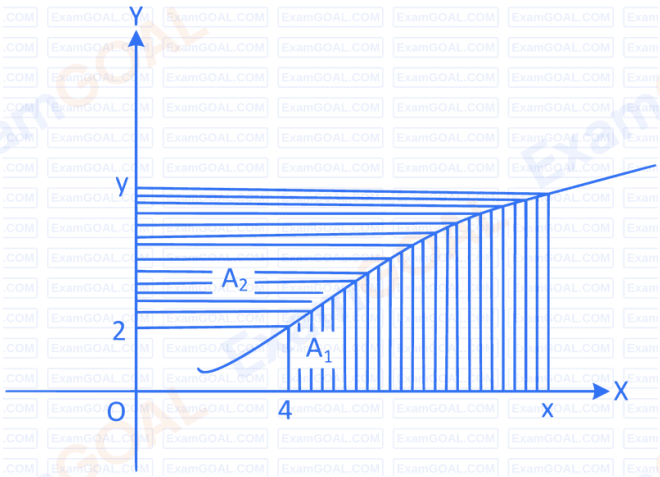
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Q30. MCQ (Single Correct Answer)

Open solution

Consider a curve $y = y(x)$ in the first quadrant as shown in the figure. Let the area A_1 is twice the area A_2. Then the normal to the curve perpendicular to the line $2x - 12y = 15$ does NOT pass through the point.



(A) (6, 21) (B) (8, 9) (C) (10, −4) (D) (12, −15)

Q31. MCQ (Single Correct Answer)

Open solution

The equations of the sides AB, BC and CA of a triangle ABC are $2x + y = 0, x + py = 39$ and $x - y = 3$ respectively and $P(2, 3)$ is its circumcentre. Then which of the following is NOT true?

(A) $(AC)^2 = 9p$ (B) $(AC)^2 + p^2 = 136$

(C) $32 < \text{area}(\Delta ABC) < 36$ (D) $34 < \text{area}(\triangle ABC) < 38$

Q32. MCQ (Single Correct Answer)

Open solution

If the length of the perpendicular drawn from the point $P(a, 4, 2)$, $a > 0$ on the line $\frac{x+1}{2} = \frac{y-3}{3} = \frac{z-1}{-1}$ is $2\sqrt{6}$ units and $Q(\alpha_1, \alpha_2, \alpha_3)$ is the image of the point P in this line, then $a + \sum_{i=1}^{3} \alpha_i$ is equal to :

(A) 7 (B) 8 (C) 12 (D) 14

Q33. MCQ (Single Correct Answer)

Open solution

A six faced die is biased such that

$3 \times P(\text{a prime number}) = 6 \times P(\text{a composite number}) = 2 \times P(1)$.

Let X be a random variable that counts the number of times one gets a perfect square on some throws of this die. If the die is thrown twice, then the mean of X is :

(A) $\frac{3}{11}$ (B) $\frac{5}{11}$ (C) $\frac{7}{11}$ (D) $\frac{8}{11}$

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Q34. Numerical

The number of functions f, from the set $A = \{x \in \mathbf{N} : x^2 - 10x + 9 \leq 0\}$ to the set $B = \{n^2 : n \in \mathbf{N}\}$ such that $f(x) \leq (x-3)^2 + 1$, for every $x \in A$, is _____.

Open solution

Q35. Numerical

Let for the 9th term in the binomial expansion of $(3 + 6x)^n$, in the increasing powers of $6x$, to be the greatest for $x = \frac{3}{2}$, the least value of n is n_0. If k is the ratio of the coefficient of x^6 to the coefficient of x^3, then $k + n_0$ is equal to :

Open solution

Q36. Numerical

A water tank has the shape of a right circular cone with axis vertical and vertex downwards. Its semi-vertical angle is $\tan^{-1}\frac{3}{4}$. Water is poured in it at a constant rate of 6 cubic meter per hour. The rate (in square meter per hour), at which the wet curved surface area of the tank is increasing, when the depth of water in the tank is 4 meters, is _____.

Open solution

Q37. Numerical

For the curve $C : (x^2 + y^2 - 3) + (x^2 - y^2 - 1)^5 = 0$, the value of $3y' - y^3y''$, at the point (α, α), $\alpha > 0$, on C, is equal to _____.

Open solution

Q38. Numerical

Let $f(x) = \min\{[x-1], [x-2], \ldots, [x-10]\}$ where [t] denotes the greatest integer $\leq t$. Then $\int_0^{10} f(x)dx + \int_0^{10} (f(x))^2\,dx + \int_0^{10} |f(x)|dx$ is equal to _____.

Open solution

Q39. Numerical

Let f be a differentiable function satisfying $f(x) = \frac{2}{\sqrt{3}}\int_0^{\sqrt{3}} f\left(\frac{\lambda^2 x}{3}\right)d\lambda, x > 0$ and $f(1) = \sqrt{3}$. If $y = f(x)$ passes through the point $(\alpha, 6)$, then α is equal to _____.

Open solution

Q40. Numerical

Let $\vec{a}, \vec{b}, \vec{c}$ be three non-coplanar vectors such that $\vec{a} \times \vec{b} = 4\vec{c}, \vec{b} \times \vec{c} = 9\vec{a}$ and $\vec{c} \times \vec{a} = \alpha\vec{b}$, $\alpha > 0$. If $|\vec{a}| + |\vec{b}| + |\vec{c}| = \frac{1}{36}$, then α is equal to _____.

Open solution

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Physics 25 questions

Q41. MCQ (Single Correct Answer)

An expression of energy density is given by $u = \frac{\alpha}{\beta}\sin\left(\frac{\alpha x}{kt}\right)$, where α, β are constants, x is displacement, k is Boltzmann constant and t is the temperature. The dimensions of β will be :

(A) $\left[\mathrm{ML^2\,T^{-2}\theta^{-1}}\right]$ (B) $\left[\mathrm{M^0\,L^2\,T^{-2}}\right]$

(C) $\left[\mathrm{M^0\,L^0\,T^0}\right]$ (D) $\left[\mathrm{M^0\,L^2\,T^0}\right]$

Open solution

Q42. MCQ (Single Correct Answer)

A body of mass $10\ \mathrm{kg}$ is projected at an angle of 45° with the horizontal. The trajectory of the body is observed to pass through a point $(20, 10)$. If T is the time of flight, then its momentum vector, at time $\mathrm{t} = \frac{\mathrm{T}}{\sqrt{2}}$, is _____.

[Take $\mathrm{g} = 10\ \mathrm{m/s^2}$]

(A) $100\hat{i} + (100\sqrt{2} - 200)\hat{j}$ (B) $100\sqrt{2}\hat{i} + (100 - 200\sqrt{2})\hat{j}$

(C) $100\hat{i} + (100 - 200\sqrt{2})\hat{j}$ (D) $100\sqrt{2}\hat{i} + (100\sqrt{2} - 200)\hat{j}$

Open solution

Q43. MCQ (Single Correct Answer)

A block of mass M slides down on a rough inclined plane with constant velocity. The angle made by the incline plane with horizontal is θ. The magnitude of the contact force will be :

(A) Mg (B) $\mathrm{Mg}\cos\theta$

(C) $\sqrt{\mathrm{Mg}\sin\theta + \mathrm{Mg}\cos\theta}$ (D) $\mathrm{Mg}\sin\theta\sqrt{1+\mu}$

Open solution

Q44. MCQ (Single Correct Answer)

A block 'A' takes 2 s to slide down a frictionless incline of 30° and length 'l', kept inside a lift going up with uniform velocity 'v'. If the incline is changed to 45°, the time taken by the block, to slide down the incline, will be approximately :

(A) 2.66 s (B) 0.83 s (C) 1.68 s (D) 0.70 s

Open solution

Q45. MCQ (Single Correct Answer)

The velocity of the bullet becomes one third after it penetrates 4 cm in a wooden block. Assuming that bullet is facing a constant resistance during its motion in the block. The bullet stops completely after travelling at (4 + x) cm inside the block. The value of x is :

(A) 2.0 (B) 1.0 (C) 0.5 (D) 1.5

Open solution

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Q46. MCQ (Single Correct Answer)

A body of mass m is projected with velocity λv_e in vertically upward direction from the surface of the earth into space. It is given that v_e is escape velocity and $\lambda < 1$. If air resistance is considered to be negligible, then the maximum height from the centre of earth, to which the body can go, will be :

(R : radius of earth)

(A) $\frac{\mathrm{R}}{1+\lambda^2}$

(B) $\frac{R}{1-\lambda^2}$

(C) $\frac{R}{1-\lambda}$

(D) $\frac{\lambda^2 \mathrm{R}}{1-\lambda^2}$

Open solution

Q47. MCQ (Single Correct Answer)

A steel wire of length $3.2\ \mathrm{m}$ $\left(\mathrm{Y_s} = 2.0 \times 10^{11}\ \mathrm{Nm^{-2}}\right)$ and a copper wire of length $4.4\ \mathrm{m}$ $\left(\mathrm{Y_c} = 1.1 \times 10^{11}\ \mathrm{Nm^{-2}}\right)$, both of radius $1.4\ \mathrm{mm}$ are connected end to end. When stretched by a load, the net elongation is found to be $1.4\ \mathrm{mm}$. The load applied, in Newton, will be: (Given $\pi = \frac{22}{7}$)

(A) 360

(B) 180

(C) 1080

(D) 154

Open solution

Q48. MCQ (Single Correct Answer)

Which statements are correct about degrees of freedom ?

(A) A molecule with n degrees of freedom has n^2 different ways of storing energy.

(B) Each degree of freedom is associated with $\frac{1}{2}$ RT average energy per mole.

(C) A monatomic gas molecule has 1 rotational degree of freedom whereas diatomic molecule has 2 rotational degrees of freedom.

(D) $\mathrm{CH_4}$ has a total of 6 degrees of freedom.

Choose the correct answer from the options given below :

(A) (B) and (C) only

(B) (B) and (D) only

(C) (A) and (B) only

(D) (C) and (D) only

Open solution

Q49. MCQ (Single Correct Answer)

A charge of $4\ \mu\mathrm{C}$ is to be divided into two. The distance between the two divided charges is constant. The magnitude of the divided charges so that the force between them is maximum, will be :

(A) $1\ \mu\mathrm{C}$ and $3\ \mu\mathrm{C}$

(B) $2\ \mu\mathrm{C}$ and $2\ \mu\mathrm{C}$

(C) 0 and $4\ \mu\ \mathrm{C}$

(D) $1.5\ \mu\mathrm{C}$ and $2.5\ \mu\mathrm{C}$

Open solution

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Q50. MCQ (Single Correct Answer)

Open solution

(A) The drift velocity of electrons decreases with the increase in the temperature of conductor.

(B) The drift velocity is inversely proportional to the area of cross-section of given conductor.

(C) The drift velocity does not depend on the applied potential difference to the conductor.

(D) The drift velocity of electron is inversely proportional to the length of the conductor.

(E) The drift velocity increases with the increase in the temperature of conductor.

Choose the correct answer from the options given below :

(A) (A) and (B) only **(B)** (A) and (D) only **(C)** (B) and (E) only **(D)** (B) and (C) only

Q51. MCQ (Single Correct Answer)

Open solution

A cyclotron is used to accelerate protons. If the operating magnetic field is $1.0\ \mathrm{T}$ and the radius of the cyclotron 'dees' is $60\ \mathrm{cm}$, the kinetic energy of the accelerated protons in MeV will be :

$[\text{use } \mathrm{m_p} = 1.6 \times 10^{-27}\ \mathrm{kg}, \mathrm{e} = 1.6 \times 10^{-19}\ \mathrm{C}\ ]$

(A) 12 **(B)** 18 **(C)** 16 **(D)** 32

Q52. MCQ (Single Correct Answer)

Open solution

A series LCR circuit has $\mathrm{L} = 0.01\ \mathrm{H}, \mathrm{R} = 10\ \Omega$ and $\mathrm{C} = 1\mu\mathrm{F}$ and it is connected to ac voltage of amplitude $(\mathrm{V_m})50\ \mathrm{V}$. At frequency 60% lower than resonant frequency, the amplitude of current will be approximately :

(A) 466 mA **(B)** 312 mA **(C)** 238 mA **(D)** 196 mA

Q53. MCQ (Single Correct Answer)

Open solution

Identify the correct statements from the following descriptions of various properties of electromagnetic waves.

(A) In a plane electromagnetic wave electric field and magnetic field must be perpendicular to each other and direction of propagation of wave should be along electric field or magnetic field.

(B) The energy in electromagnetic wave is divided equally between electric and magnetic fields.

(C) Both electric field and magnetic field are parallel to each other and perpendicular to the direction of propagation of wave.

(D) The electric field, magnetic field and direction of propagation of wave must be perpendicular to each other.

(E) The ratio of amplitude of magnetic field to the amplitude of electric field is equal to speed of light.

Choose the most appropriate answer from the options given below :

(A) (D) only **(B)** (B) and (D) only

(C) (B), (C) and (E) only **(D)** (A), (B) and (E) only

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Q54. MCQ (Single Correct Answer)

Open solution

Two coherent sources of light interfere. The intensity ratio of two sources is $1:4$. For this interference pattern if the value of $\frac{I_{max}+I_{min}}{I_{max}-I_{min}}$ is equal to $\frac{2\alpha+1}{\beta+3}$, then $\frac{\alpha}{\beta}$ will be :

(A) 1.5 (B) 2 (C) 0.5 (D) 1

Q55. MCQ (Single Correct Answer)

Open solution

With reference to the observations in photo-electric effect, identify the correct statements from below :

(A) The square of maximum velocity of photoelectrons varies linearly with frequency of incident light.

(B) The value of saturation current increases on moving the source of light away from the metal surface.

(C) The maximum kinetic energy of photo-electrons decreases on decreasing the power of LED (light emitting diode) source of light.

(D) The immediate emission of photo-electrons out of metal surface can not be explained by particle nature of light/electromagnetic waves.

(E) Existence of threshold wavelength can not be explained by wave nature of light/ electromagnetic waves.

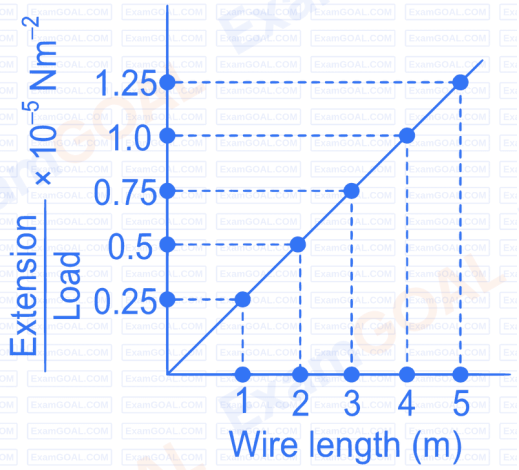
Choose the correct answer from the options given below :

(A) (A) and (B) only **(B)** (A) and (E) only **(C)** (C) and (E) only **(D)** (D) and (E) only

Q56. Numerical

Open solution

In an experiment to determine the Young's modulus, steel wires of five different lengths $(1, 2, 3, 4,$ and $5\ \mathrm{m})$ but of same cross section $\left(2\ \mathrm{mm}^2\right)$ were taken and curves between extension and load were obtained. The slope (extension/load) of the curves were plotted with the wire length and the following graph is obtained. If the Young's modulus of given steel wires is $x \times 10^{11}\ \mathrm{Nm}^{-2}$, then the value of x is _____.

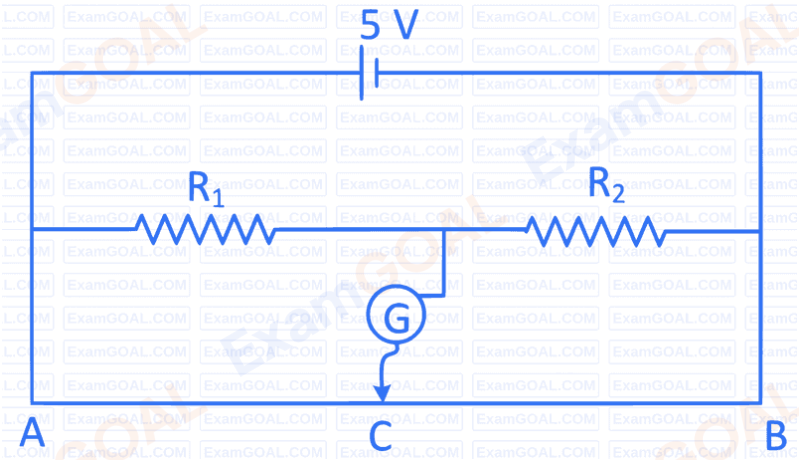


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Q57. Numerical

In the given figure of meter bridge experiment, the balancing length AC corresponding to null deflection of the galvanometer is $40\ \mathrm{cm}$. The balancing length, if the radius of the wire AB is doubled, will be _____ cm.

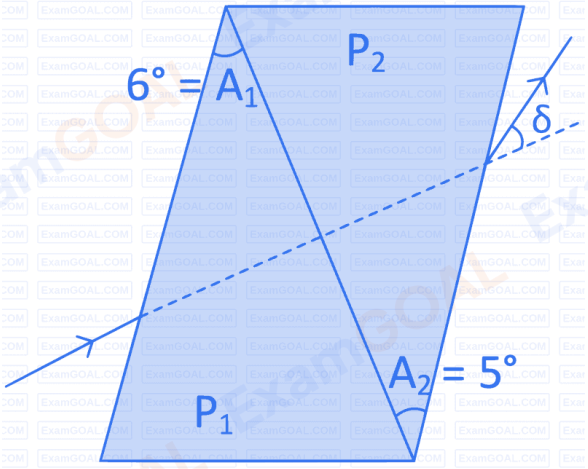
Open solution



Q58. Numerical

A thin prism of angle 6° and refractive index for yellow light $(\mathrm{n_Y})1.5$ is combined with another prism of angle 5° and $\mathrm{n_Y} = 1.55$. The combination produces no dispersion. The net average deviation (δ) produced by the combination is $\left(\frac{1}{x}\right)^\circ$. The value of x is _____.

Open solution



Q59. Numerical

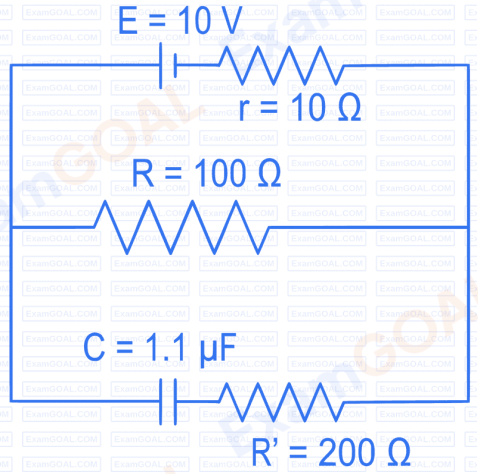
A conducting circular loop is placed in $X - Y$ plane in presence of magnetic field $\overrightarrow{\mathrm{B}} = \left(3\mathrm{t}^3\,\hat{j} + 3\mathrm{t}^2\,\hat{k}\right)$ in SI unit. If the radius of the loop is $1\ \mathrm{m}$, the induced emf in the loop, at time, $\mathrm{t} = 2\ \mathrm{s}$ is $\mathrm{n}\pi\ \mathrm{V}$. The value of n is _____.

Open solution

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Q60. Numerical

As shown in the figure, in steady state, the charge stored in the capacitor is _____ $\times 10^{-6}$ C.



Open solution

Q61. Numerical

A parallel plate capacitor with width $4\ \text{cm}$, length $8\ \text{cm}$ and separation between the plates of $4\ \text{mm}$ is connected to a battery of $20\ \text{V}$. A dielectric slab of dielectric constant 5 having length $1\ \text{cm}$, width $4\ \text{cm}$ and thickness $4\ \text{mm}$ is inserted between the plates of parallel plate capacitor. The electrostatic energy of this system will be _____ ϵ_0 J. (Where ϵ_0 is the permittivity of free space)

Open solution

Q62. Numerical

A wire of length 30 cm, stretched between rigid supports, has it's n^{th} and $(n + 1)^{th}$ harmonics at 400 Hz and 450 Hz, respectively. If tension in the string is 2700 N, it's linear mass density is _____ kg/m.

Open solution

Q63. Numerical

A spherical soap bubble of radius 3 cm is formed inside another spherical soap bubble of radius 6 cm. If the internal pressure of the smaller bubble of radius 3 cm in the above system is equal to the internal pressure of another single soap bubble of radius r cm. The value of r is _____.

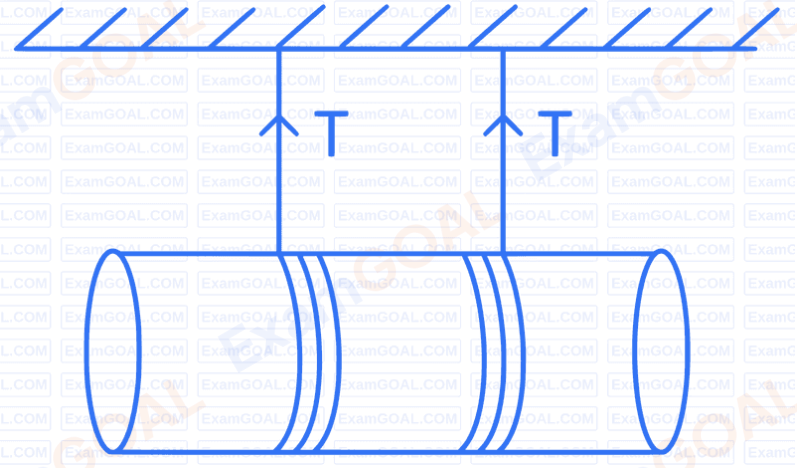
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Q64. Numerical

A solid cylinder is suspended symmetrically through two massless strings, as shown in the figure. The distance from the initial rest position, the cylinder should be unbinding the strings to achieve a speed of $4\ \text{ms}^{-1}$, is _____ cm. (take g = $10\ \text{ms}^{-2}$)

Open solution

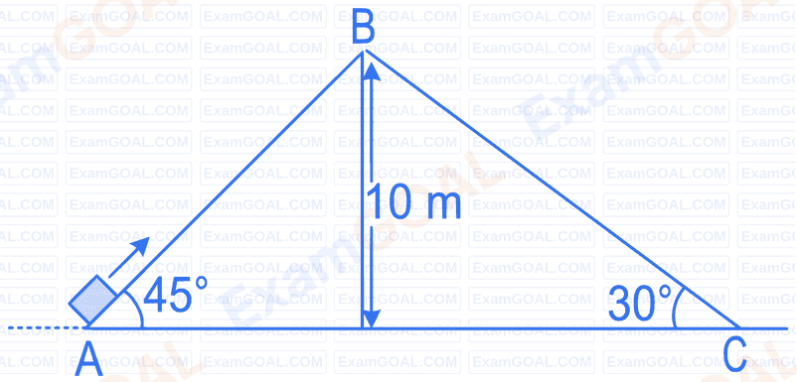


Q65. Numerical

Two inclined planes are placed as shown in figure. A block is projected from the Point A of inclined plane AB along its surface with a velocity just sufficient to carry it to the top Point B at a height 10 m. After reaching the Point B the block slides down on inclined plane BC. Time it takes to reach to the point C from point A is $t(\sqrt{2}+1)$ s. The value of t is _____.

(use $g = 10\ \text{m/s}^2$)

Open solution



Out of syllabus 25 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

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Q66. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Dissolved substances can be removed from a colloidal solution by diffusion through a parchment paper.

Reason (R) : Particles in a true solution cannot pass through parchment paper but the colloidal particles can pass through the parchment paper.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(B) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(C) (A) is correct but (R) is not correct

(D) (A) is not correct but (R) is correct

Q67. MCQ (Single Correct Answer) · **Chemistry**

Open solution

An element A of group 1 shows similarity to an element B belonging to group 2. If A has maximum hydration enthalpy in group 1 then B is :

(A) Mg **(B)** Be **(C)** Ca **(D)** Sr

Q68. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Boron is unable to form BF_6^{3-}.

Reason (R) : Size of B is very small.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both (A) and (R) are true and (R) is the correct explanation of (A)

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(C) (A) is true but (R) is false

(D) (A) is false but (R) is true

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Q69. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Given below are two statements:

Statement I : The non bio-degradable fly ash and slag from steel industry can be used by cement industry.

Statement II : The fuel obtained from plastic waste is lead free.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both Statement I and Statement II are correct

(B) Both Statement I and Statement II are incorrect

(C) Statement I is correct but Statement II is incorrect

(D) Statement I is incorrect but Statement II is correct

Q70. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Match List - I with List - II.

	List - I (Polymer)		**List - II (Monomer)**
(A)	Neoprene	(I)	Acrylonitrile
(B)	Teflon	(II)	Chloroprene
(C)	Acrilan	(III)	Tetrafluoroethene
(D)	Natural rubber	(IV)	Isoprene

Choose the correct answer from the options given below :

(A) $(A)-(II),(B)-(III),(C)-(I),(D)-(IV)$

(B) $(A)-(II),(B)-(I),(C)-(III),(D)-(IV)$

(C) $(A)-(II),(B)-(I),(C)-(IV),(D)-(III)$

(D) $(A)-(I),(B)-(II),(C)-(III),(D)-(IV)$

Q71. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Which of the following enhances the lathering property of soap?

(A) Sodium stearate

(B) Sodium carbonate

(C) Sodium rosinate

(D) Trisodium phosphate

Q72. Numerical · **Chemistry**

Open solution

For a real gas at $25^{\circ}C$ temperature and high pressure (99 bar) the value of compressibility factor is 2, so the value of van der Waals constant 'b' should be _____ $\times 10^{-2}\ L\ mol^{-1}$ (Nearest integer)

(Given $R = 0.083\ L\ bar\ K^{-1}\ mol^{-1}$)

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Q73. Numerical · **Chemistry**

Among the following ores Bauxite, Siderite, Cuprite, Calamine, Haematite, Kaolinite, Malachite, Magnetite, Sphalerite, Limonite, Cryolite, the number of principal ores of iron is _____.

Open solution

Q74. Numerical · **Chemistry**

The oxidation state of manganese in the product obtained in a reaction of potassium permanganate and hydrogen peroxide in basic medium is _____.

Open solution

Q75. MCQ (Single Correct Answer) · **Mathematics**

Let $A = \begin{pmatrix} 4 & -2 \\ \alpha & \beta \end{pmatrix}$.

If $\mathrm{A}^2 + \gamma\mathrm{A} + 18\mathrm{I} = \mathrm{O}$, then $\det(\mathrm{A})$ is equal to _____.

(A) -18 (B) 18 (C) -50 (D) 50

Open solution

Q76. MCQ (Single Correct Answer) · **Mathematics**

A circle C_1 passes through the origin O and has diameter 4 on the positive x-axis. The line $y = 2x$ gives a chord OA of circle C_1. Let C_2 be the circle with OA as a diameter. If the tangent to C_2 at the point A meets the x-axis at P and y-axis at Q, then $\mathrm{QA} : \mathrm{AP}$ is equal to :

(A) 1 : 4 (B) 1 : 5 (C) 2 : 5 (D) 1 : 3

Open solution

Q77. MCQ (Single Correct Answer) · **Mathematics**

If the length of the latus rectum of a parabola, whose focus is (a, a) and the tangent at its vertex is $x + y = a$, is 16, then $|a|$ is equal to :

(A) $2\sqrt{2}$ (B) $2\sqrt{3}$ (C) $4\sqrt{2}$ (D) 4

Open solution

Q78. MCQ (Single Correct Answer) · **Mathematics**

If the line of intersection of the planes $ax + by = 3$ and $ax + by + cz = 0$, $a > 0$ makes an angle 30° with the plane $y - z + 2 = 0$, then the direction cosines of the line are :

(A) $\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0$ (B) $\frac{1}{\sqrt{2}}, \pm\frac{1}{\sqrt{2}}, 0$

(C) $\frac{1}{\sqrt{5}}, -\frac{2}{\sqrt{5}}, 0$ (D) $\frac{1}{2}, -\frac{\sqrt{3}}{2}, 0$

Open solution

Q79. MCQ (Single Correct Answer) · **Mathematics**

Let X have a binomial distribution B(n, p) such that the sum and the product of the mean and variance of X are 24 and 128 respectively. If $P(X > n - 3) = \frac{k}{2^n}$, then k is equal to :

(A) 528 (B) 529 (C) 629 (D) 630

Open solution

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Q80. MCQ (Single Correct Answer) · **Mathematics**

The angle of elevation of the top P of a vertical tower PQ of height 10 from a point A on the horizontal ground is 45°. Let R be a point on AQ and from a point B, vertically above R, the angle of elevation of P is 60°. If $\angle BAQ = 30^\circ$, $AB = d$ and the area of the trapezium PQRB is α, then the ordered pair (d, α) is :

(A) $(10(\sqrt{3}-1), 25)$

(B) $\left(10(\sqrt{3}-1), \frac{25}{2}\right)$

(C) $(10(\sqrt{3}+1), 25)$

(D) $\left(10(\sqrt{3}+1), \frac{25}{2}\right)$

Open solution

Q81. MCQ (Single Correct Answer) · **Mathematics**

Let $S = \left\{\theta \in \left(0, \frac{\pi}{2}\right) : \sum_{m=1}^{9} \sec\left(\theta + (m-1)\frac{\pi}{6}\right) \sec\left(\theta + \frac{m\pi}{6}\right) = -\frac{8}{\sqrt{3}}\right\}$. Then

(A) $S = \left\{\frac{\pi}{12}\right\}$

(B) $S = \left\{\frac{2\pi}{3}\right\}$

(C) $\sum_{\theta \in S} \theta = \frac{\pi}{2}$

(D) $\sum_{\theta \in S} \theta = \frac{3\pi}{4}$

Open solution

Q82. MCQ (Single Correct Answer) · **Mathematics**

If the truth value of the statement $(P \wedge (\sim R)) \rightarrow ((\sim R) \wedge Q)$ is F, then the truth value of which of the following is F ?

(A) $P \vee Q \rightarrow \sim R$

(B) $R \vee Q \rightarrow \sim P$

(C) $\sim (P \vee Q) \rightarrow \sim R$

(D) $\sim (R \vee Q) \rightarrow \sim P$

Open solution

Q83. Numerical · **Mathematics**

Consider a matrix $A = \begin{bmatrix} \alpha & \beta & \gamma \\ \alpha^2 & \beta^2 & \gamma^2 \\ \beta+\gamma & \gamma+\alpha & \alpha+\beta \end{bmatrix}$, where α, β, γ are three distinct natural numbers.

If $\frac{\det(\text{adj}(\text{adj}(\text{adj}(\text{adj}\, A))))}{(\alpha-\beta)^{16}(\beta-\gamma)^{16}(\gamma-\alpha)^{16}} = 2^{32} \times 3^{16}$, then the number of such 3 - tuples (α, β, γ) is _____.

Open solution

Q84. Numerical · **Mathematics**

$\frac{2^3-1^3}{1\times7} + \frac{4^3-3^3+2^3-1^3}{2\times11} + \frac{6^3-5^3+4^3-3^3+2^3-1^3}{3\times15} + \cdots + \frac{30^3-29^3+28^3-27^3+\ldots+2^3-1^3}{15\times63}$ is equal to _____.

Open solution

Q85. Numerical · **Mathematics**

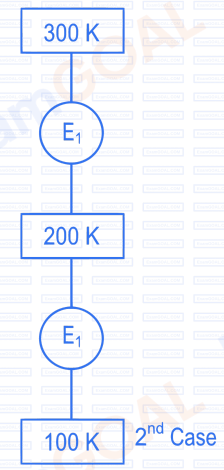
A common tangent T to the curves $C_1 : \frac{x^2}{4} + \frac{y^2}{9} = 1$ and $C_2 : \frac{x^2}{42} - \frac{y^2}{143} = 1$ does not pass through the fourth quadrant. If T touches C_1 at (x_1, y_1) and C_2 at (x_2, y_2), then $|2x_1 + x_2|$ is equal to _____.

Open solution

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Q86. MCQ (Single Correct Answer) · **Physics**

In 1st case, Carnot engine operates between temperatures $300\ \mathrm{K}$ and $100\ \mathrm{K}$. In 2nd case, as shown in the figure, a combination of two engines is used. The efficiency of this combination (in 2nd case) will be :



Open solution

(A) same as the 1st case.

(B) always greater than the 1st case.

(C) always less than the 1st case.

(D) may increase or decrease with respect to the 1st case.

Q87. MCQ (Single Correct Answer) · **Physics**

A compass needle of oscillation magnetometer oscillates 20 times per minute at a place P of $\mathrm{dip}\ 30^{\circ}$. The number of oscillations per minute become 10 at another place Q of 60° dip. The ratio of the total magnetic field at the two places $(B_Q : B_P)$ is :

Open solution

(A) $\sqrt{3} : 4$ (B) $4 : \sqrt{3}$ (C) $\sqrt{3} : 2$ (D) $2 : \sqrt{3}$

Q88. MCQ (Single Correct Answer) · **Physics**

The activity of a radioactive material is 6.4×10^{-4} curie. Its half life is 5 days. The activity will become 5×10^{-6} curie after :

Open solution

(A) 7 days (B) 15 days (C) 25 days (D) 35 days

Q89. MCQ (Single Correct Answer) · **Physics**

For a constant collector-emitter voltage of $8\ \mathrm{V}$, the collector current of a transistor reached to the value of $6\ \mathrm{mA}$ from $4\ \mathrm{mA}$, whereas base current changed from $20\ \mu\mathrm{A}$ to $25\ \mu\mathrm{A}$ value. If transistor is in active state, small signal current gain (current amplification factor) will be :

Open solution

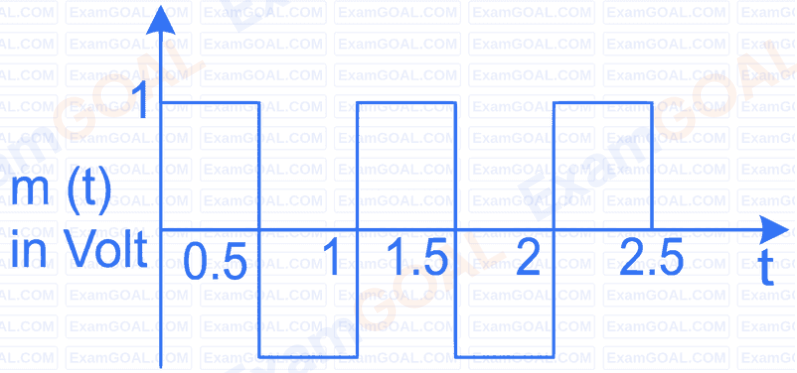
(A) 240 (B) 400 (C) 0.0025 (D) 200

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Q90. MCQ (Single Correct Answer) · **Physics**

Open solution

A square wave of the modulating signal is shown in the figure. The carrier wave is given by $C(t) = 5\sin(8\pi t)$ Volt. The modulation index is :



(A) 0.2 (B) 0.1 (C) 0.3 (D) 0.4

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Answer Key

Chemistry

1. **A**	2. **C**	3. **A**	4. **B**	5. **B**
6. **D**	7. **A**	8. **C**	9. **B**	10. **A**
11. **D**	12. **A**	13. **D**	14. **D**	15. **1**
16. **35**	17. **6**	18. **165**	19. **6**	20. **2**
21. **530**				

Mathematics

22. **C**	23. **C**	24. **B**	25. **B**	26. **D**
27. **A**	28. **D**	29. **B**	30. **C**	31. **D**
32. **B**	33. **D**	34. **1440**	35. **24**	36. **5**
37. **16**	38. **385**	39. **12**	40. **36**	

Physics

41. **D**	42. **D**	43. **A**	44. **C**	45. **C**
46. **B**	47. **D**	48. **B**	49. **B**	50. **B**
51. **B**	52. **C**	53. **B**	54. **B**	55. **B**
56. **2**	57. **40**	58. **4**	59. **12**	60. **10**
61. **240**	62. **3**	63. **2**	64. **120**	65. **2**

Out of syllabus

66. **C**	67. **A**	68. **B**	69. **A**	70. **A**
71. **C**	72. **25**	73. **4**	74. **4**	75. **B**
76. **A**	77. **C**	78. **B**	79. **B**	80. **A**
81. **C**	82. **D**	83. **42**	84. **120**	85. **20**
86. **A**	87. **A**	88. **D**	89. **B**	90. **A**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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